

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India 09/2021 – 01/2025 (3.5 Years)

- Designed and delivered **geotechnical analysis** for **foundations** and **retaining structures** on major international commissions, completing analysis 2 weeks ahead of schedule.
- Prepared detailed **design reports**, **technical specifications**, and **desk studies** for commercial developments, ensuring compliance with project requirements and industry standards.
- Planned and supervised **geotechnical ground investigations**, interpreting data to provide critical technical support during the construction phase, reducing on-site query resolution time by **15%**.
- Integrated **digital tools** and **automation** into project workflows for **numerical analysis**, enhancing design efficiency and accuracy by an estimated **20%**.
- Managed project timelines and resources for multiple **geotechnical engineering** assignments, delivering projects within budget and scope.
- Collaborated with cross-functional teams to ensure seamless integration of geotechnical designs with overall project objectives, contributing to the successful delivery of infrastructure projects.

SKILLS

Geotechnical Engineering – Geotechnical Design, Foundations, Retaining Structures, Slope Stability, Earthworks, Ground Investigations, Numerical Analysis, GIS

Engineering Software – AutoCAD, Civil 3D, GIS Software, PLAXIS (or similar numerical modelling)

Project Management – Planning, Scheduling, Resource Allocation, Technical Specifications, Design Reports, Site Supervision, Client Communication

Data Analysis – Data Interpretation, Statistical Analysis, Python (pandas, NumPy), SQL

Cloud & Automation – AWS, Docker, Jenkins, Python Scripting

Development Tools – Git, Jira, Confluence

Core Technologies – Java, Spring Boot, REST APIs, PostgreSQL, Linux

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin 02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India 01/2021

PROJECTS

Automated Geotechnical Analysis Platform (HCL Technologies)

- Developed a Python-based platform to automate routine **geotechnical analysis** tasks, including **slope stability** and **foundation bearing capacity** calculations, reducing manual analysis time by **40%**.
- Integrated **SQL** databases (PostgreSQL) for storing and retrieving historical ground investigation data, enabling faster site-specific analysis and **desk study** report generation.

- Designed and implemented REST APIs using **FastAPI** to allow seamless data input and output for the analysis engine, facilitating integration with broader project management tools.

Payments Operations Dashboard Enhancement (HCL Technologies)

- Enhanced a **payments operations dashboard** by developing new Java Spring Boot microservices to process and monitor transaction flows, improving the visibility of failed payment messages by **30%**.
- Implemented real-time alerting mechanisms using **Splunk** and **Dynatrace** integrations, reducing incident response times for critical payment issues by **25%**.
- Contributed to maintaining **audit trails** and ensuring **GDPR**-aware data handling within the financial-services production environment.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)